

Student Number:.....



Mathematics Extension 2

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

- Attempt questions 1-8
- All questions are of equal value

[illegible]

Question 1 (15 marks)**Start a new sheet of writing paper.****Marks**

a) Find

$$\int_0^{\frac{\pi}{4}} \tan^4 x \sec^2 x \, dx.$$

2

b)

Find $\int \frac{dx}{\sqrt{x^2 - 6x + 8}} dx$

2

c)

Use partial fractions to show

3

$$\int_2^5 \frac{2x+2}{(x-1)(2x-1)} dx = \log_e \left(\frac{256}{27} \right).$$

d)

Find $\int \sin(\log_e x) \, dx.$

4

e)

Use the substitution $t = \tan \frac{x}{2}$ to find**4**

$$\int_0^{\frac{\pi}{2}} \frac{dx}{(3 \cos x + 4 \sin x + 5)}.$$

End of Question 1

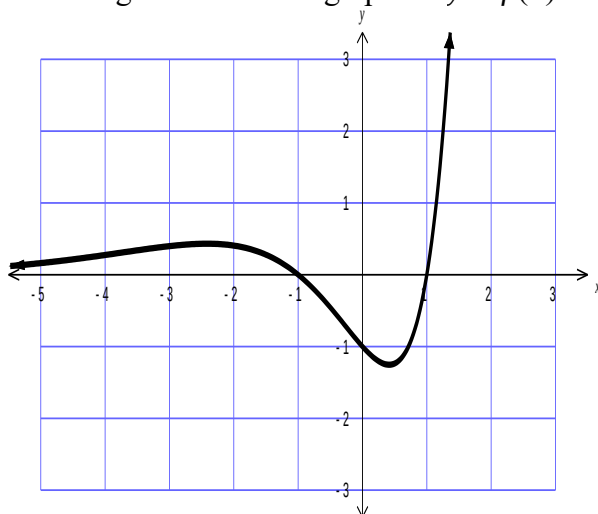
Question 2 (15 marks)**Start a new sheet of writing paper.****Marks**

- a) If $z = \sqrt{3} + i$ and $w = 1 - i$
- i) Write $\frac{z}{w}$ in the form $a + ib$ where a and b are real. 1
- ii) Write $\frac{z}{w}$ in mod-arg form. 2
- iii) Hence, or otherwise, show that $\cos \frac{5\pi}{12} = \frac{\sqrt{6} - \sqrt{2}}{4}$ 2
- iv) Express $\left(\frac{z}{w}\right)^{12}$ in the form $a + ib$ where a and b are real. 2
- b) The points Z , W and O on the Argand diagram represent the complex numbers z , w and o respectively. If $z = 3 + i$ and $o = 0 + 0i$. Find the complex number w , in $a + ib$ form where a and b are real, if $\triangle OZW$ in anti-clockwise order, is right-angled at Z and the distance from Z to W is twice the distance from O to Z . 2
- c) The point P on the Argand diagram represents the complex number $z = x + iy$ which satisfies $(z)^2 = 2 - (\bar{z})^2$. Find the equation of the locus of P in terms of x and y . What type of curve is this locus? 3
- d) If z is a complex number such that $z = r(\cos \theta + i \sin \theta)$, where r is real, show that $\arg(z + r) = \frac{1}{2} \theta$. 3

End of Question 2

Question 3 (15 marks)**Start a new sheet of writing paper.****Marks**

- a) The diagram shows the graph of $y = f(x)$.



Draw separate one-third page sketches of the graphs of the following:

- | | | |
|------|----------------------|---|
| i) | $y = \frac{1}{f(x)}$ | 1 |
| ii) | $ y = f(x)$ | 1 |
| iii) | $y = [f(x)]^2$ | 2 |
| iv) | $y = \sqrt{f(x)}$ | 2 |
| v) | $y = x(f(x))$ | 2 |
-
- | | | | |
|----|------|---|---|
| b) | i) | Express the complex number $1+i$ in the form $r(\cos \theta + i \sin \theta)$. | 1 |
| | ii) | Hence prove that $(1+i)^n + (1-i)^n = 2(2^{\frac{n}{2}} \cos \frac{n\pi}{4})$ where n is a positive integer. | 3 |
| | iii) | If $(1+x)^n = p_0 + p_1x + p_2x^2 + \dots + p_nx^n$, prove that
$p_0 - p_2 + p_4 - \dots = 2^{\frac{n}{2}} \cos \frac{n\pi}{4} \text{ and } p_1 - p_3 + p_5 - \dots = 2^{\frac{n}{2}} \sin \frac{n\pi}{4}.$ | 3 |

End of Question 3

Question 4 (15 marks)**Start a new sheet of writing paper.****Marks**

- a) Consider the hyperbola H with equation $\frac{(x-1)^2}{16} - \frac{(y+3)^2}{25} = 1$
- i) Find the centre, the eccentricity and the co-ordinates of the foci of H . **3**
- ii) Write down the equations of the directrices and the asymptotes of H . **3**
- iii) Sketch H showing all of the above features. **1**
- b) Using the focus-directrix definition of the ellipse, centred at the origin, prove that the sum of the focal lengths is constant. **2**
- c) Given the hyperbola $x^2 - y^2 = a^2$
- i) Show that $(a \sec \theta, a \tan \theta)$ are the parametric coordinates of a point on the hyperbola. **1**
- ii) Show that the equation of the tangent to $x^2 - y^2 = a^2$ at $(a \sec \theta, a \tan \theta)$ is $x - y \sin \theta = a \cos \theta$. **2**
- iii) Prove that the area of the triangle bounded by a tangent and the asymptotes is a constant. **3**

End of Question 4

Question 5 (15 marks) **Start a new sheet of writing paper.****Marks**

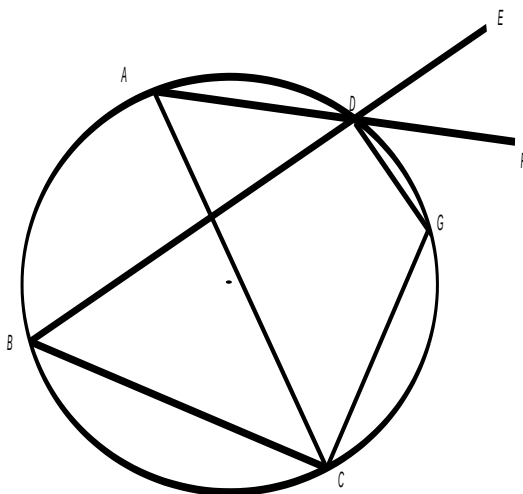
- a) Consider $P(x) = x^4 - 6x^2 - 8x - 3$.
- i) Given that $P(x)$ has a zero of multiplicity 3, express $P(x)$ as a product of linear factors. **2**
- ii) Sketch the graph of $P(x)$. **2**
- b) $P(x)$ is a polynomial of the form $P(x) = ax^3 + bx^2 + cx + d$, where a, b, c and d are real. **4**
 $P(x)$ has roots of 5 and i and when divided by $(x-2)$ the remainder is 15.
Find $P(x)$.
- c) i) Show that $(1 - \sqrt{x})^{n-1} - (1 - \sqrt{x})^n = (1 - \sqrt{x})^{n-1} \sqrt{x}$. **1**
- ii) If $I_n = \int_0^1 (1 - \sqrt{x})^n dx$ for $n \geq 0$ show that $I_n = \frac{n}{n+2} I_{n-1}$ for $n \geq 1$. **2**
- iii) Hence show that $\frac{1}{I_n} = {}^{n+2}C_n$ for $n \geq 0$. **2**
- d) If $a > 0, b > 0, c > 0, d > 0$, show that $\frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a} \geq 4$ **2**

End of Question 5

Question 6 (15 marks)**Start a new sheet of writing paper.****Marks**

- a) i) Prove that $\sin 5\theta = 16\sin^5 \theta - 20\sin^3 \theta + 5\sin \theta$ **3**
- ii) Show that roots of the equation $16x^4 - 20x^2 + 5 = 0$ are $x = \sin \frac{k\pi}{5}$, **2**
where $k = 1, 2, 3, 4$.
- iii) Construct an equation whose roots are each 1 greater than those of $16x^4 - 20x^2 + 5 = 0$ **2**
- iv) Hence or otherwise find the exact value of $\sum_{k=1}^4 \frac{1}{1 + \sin \frac{k\pi}{5}}$ **2**
- b) Find the equation of the tangent to the curve $5x^2 - 6xy + y^2 - 2x + 4y - 3 = 0$ at the point $(1, 2)$. **3**

c)



AC bisects $\angle BCG$
 ADF and BDE are straight lines.

Prove that FD bisects $\angle EDG$.

3**End of Question 6**

Question 7 (15 marks)**Start a new sheet of writing paper.****Marks**

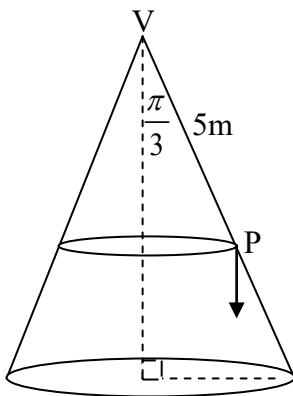
- a) The n^{th} derivative of $f(x)$ is $\frac{d^n}{dx^n} f(x) = \frac{d^{n-1}}{dx^{n-1}} \left[\frac{d}{dx} f(x) \right]$.
- i) Show that $\frac{d^n}{dx^n} (x^n) = n!$ 2
- ii) Prove, by mathematical induction, that for all positive integers, n 3
$$\frac{d^n}{dx^n} (x^n \ln x) = n! \left(\ln x + \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} \right)$$
- b) A railway line has been constructed around a circular curve of radius 500m. The distance between the rails is 1.5m and the outside rail is 0.1m above the inside rail.
- i) Draw a diagram showing all forces on the train. 1
- ii) Show that $\tan \theta = \frac{v^2}{gr}$, given that there is no sideways force on the wheels for a train on this curve. 2
- iii) Find the optimal speed for the train around this curve. (Take $g = 9.8 \text{ m/s}^2$) 1
- c) A particle of mass m projected vertically upwards with initial speed $u \text{ m/s}$ experiences a resistance of magnitude Kmv Newtons when the speed is $v \text{ m/s}$ where K is a positive constant. After T seconds the particle attains its maximum height h . Let the acceleration due to gravity be $g \text{ m/s}^2$.
- i) Show that the acceleration of the particle is given by $\ddot{x} = -(g + Kv)$ where x is the height of the particle t seconds after the launch. 1
- ii) Prove that T is given by $T = \frac{1}{K} \log_e \left(\frac{g + Ku}{g} \right)$ seconds. 2
- iii) Prove that h is given by $h = \frac{u - gT}{K}$ metres. 3

End of Question 7

Question 8 (15 marks)**Start a new sheet of writing paper.****Marks**

- a) A particle is projected from the origin with initial velocity U to pass through a point (a,b) .
- i) Show that the Cartesian equation of the motion of the particle is given by $y = \frac{-gx^2}{2U^2} \sec^2 \alpha + x \tan \alpha$. You must DERIVE all equations of motion. **3**
- ii) Prove that there are two possible trajectories if: $(U^2 - gb)^2 > g^2(a^2 + b^2)$ **3**

- b) A circular cone of semi-vertical angle $\frac{\pi}{3}$ is fixed with its vertex upwards.
- A particle P of mass m kg is attached to the vertex at V by a light inextensible string of length $5m$. The particle P rotates with uniform angular velocity ω rad/sec in a horizontal circle whose centre is vertically below V , on the outside surface of the cone and in contact with it. Let T be the tension in the string, N the normal reaction force and mg the gravitational force at P .



- i) Resolve the forces on P in the horizontal and vertical directions. **3**
- ii) Show that $T = \frac{m}{4}(2g + 15\omega^2)$ and find a similar expression for N . **4**
- iii) Show that for the particle to remain in uniform circular motion on the surface of the cone, then $\omega^2 < \frac{2g}{5}$, where g is the acceleration due to gravity. **2**

End of Examination

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

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Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2010
Course	Ext. 2	Name of task/exam	Trial Exam

Question 1

a) $\int \tan^4 x \sec^2 x \, dx$

$$u = \tan x$$

$$du = \sec^2 x \, dx$$

$$= \int u^4 \, du$$

$$= \frac{u^5}{5} + C$$

$$= \frac{\tan^5 x}{5} + C$$

b) $\int \frac{dx}{\sqrt{x^2 - 6x + 8}}$

$$= \int \frac{dx}{\sqrt{(x-3)^2 - 1}}$$

$$= \ln \left(x-3 + \sqrt{x^2 - 6x + 8} \right) + C$$

c) $\int \frac{2x+2}{(x-1)(2x-1)} \, dx$

$$\frac{2x+2}{(x-1)(2x-1)} = \frac{A}{x-1} + \frac{B}{2x-1}$$

$$= \frac{4}{x-1} - \frac{6}{2x-1}$$

$$\therefore \int \frac{2x+2}{(x-1)(2x-1)} \, dx = \int \frac{4}{x-1} - \frac{6}{2x-1} \, dx$$

$$= 4 \ln(x-1) - 3 \ln(2x-1)$$

$$= \left[\ln \left(\frac{x-1}{(2x-1)^3} \right) \right]_2^5$$

$$= \ln \frac{4^4}{9^3} - \ln \left(\frac{1}{3^3} \right)$$

$$= \ln \frac{4^4}{3^6} \times \frac{3^3}{1}$$

$$= \ln \frac{4^4}{3^3}$$

$$= \ln \left(\frac{256}{27} \right)$$

e) $\int_0^{\pi/2} \frac{dx}{3 \cos x + 4 \sin x + 5}$

$$t = \tan \frac{x}{2}$$

$$dx = \frac{2 \sec^2 \frac{x}{2}}{2}$$

$$dx = \frac{2}{\sec^2 \frac{x}{2}}$$

$$= \frac{2}{1+t^2}$$

$$x=0, t=0$$

$$x=\pi/2, t=1$$

Solutions for exams and assessment tasks

Academic Year	Yr 12	Calendar Year	2010
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$$= \int_0^1 \frac{\frac{2}{1+t^2} dt}{3\left(\frac{1-t^2}{1+t^2}\right) + 4\left(\frac{2t}{1+t^2}\right) + 5}$$

$$= \int_0^1 \frac{2}{3 - 3t^2 + 8t + 5 + 5t^2} dt$$

$$= \int_0^1 \frac{2}{2t^2 + 8t + 8} dt$$

$$= \int_0^1 \frac{1}{t^2 + 4t + 4} dt$$

$$= \int_0^1 \frac{1}{(t+2)^2} dt$$

$$= \left[\frac{-1}{t+2} \right]_0^1$$

$$= \left[-\frac{1}{3} + \frac{1}{2} \right]$$

$$= \frac{1}{6}$$

$$d) \int \sin(\log x) dx$$

$$u = \sin(\log x) \quad v = x$$

$$du = \frac{1}{x} \cos(\log x) \quad dv = 1$$

$$= x \sin(\log x) - \int x \cdot \frac{1}{x} \cos(\log x) dx$$

$$= x \sin(\log x) - \int \cos(\log x) dx$$

$$u = \cos(\log x) \quad v = x$$

$$du = -\frac{1}{x} \sin(\log x) \quad dv = 1$$

$$= x \sin(\log x) - x \cos(\log x)$$

$$+ \int x \cdot \frac{1}{x} \sin(\log x) dx$$

$$= x \sin(\log x) - x \cos(\log x)$$

$$- \int \sin(\log x) dx$$

$$\therefore 2 \int \sin(\log x)$$

$$= x \sin(\log x) - x \cos(\log x)$$

$$\therefore \int \sin(\ln x) = \frac{x}{2} \left(\sin \right.$$

$$\left. - \cos(\log_e x) \right)$$

Academic Year	Yr 12	Calendar Year	2010
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Question 2 :

a) $z = \sqrt{3} + i$ $w = 1 - i$

$$\begin{aligned} \text{i) } \frac{z}{w} &= \frac{\sqrt{3} + i}{1 - i} \\ &= \frac{\sqrt{3} + i}{1 - i} \times \frac{1 + i}{1 + i} \\ &= \frac{\sqrt{3} + \sqrt{3}i + i - 1}{1 + 1} \\ &= \frac{(\sqrt{3} - 1) + i(\sqrt{3} + 1)}{2} \end{aligned}$$

ii) $|\sqrt{3} + i| = \sqrt{(\sqrt{3})^2 + (1)^2}$

$$|z| = 2$$

$$\arg z = \frac{\pi}{6}$$

$$\therefore z = 2 \operatorname{cis} \frac{\pi}{6}$$

$$|1 - i| = \sqrt{(1)^2 + (-1)^2}$$

$$|w| = \sqrt{2}$$

$$\arg w = -\frac{\pi}{4}$$

$$\therefore w = \sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)$$

$$\therefore \frac{z}{w} = \frac{2 \operatorname{cis} \frac{\pi}{6}}{\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)}$$

$$= \sqrt{2} \operatorname{cis} \left(\frac{\pi}{6} - -\frac{\pi}{4}\right)$$

$$= \sqrt{2} \operatorname{cis} \frac{5\pi}{12}$$

iii) $\therefore \sqrt{2} \operatorname{cis} \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2} + i \frac{(\sqrt{3}+1)}{2}$

$\therefore$ equating real parts

$$\sqrt{2} \cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2}$$

$$\therefore \cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

$$= \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$\therefore \cos \frac{5\pi}{12} = \frac{\sqrt{6}-\sqrt{2}}{4}$$

iv) $\left(\frac{z}{w}\right)^{12} = \left(\frac{2 \operatorname{cis} \frac{\pi}{6}}{\sqrt{2} \operatorname{cis} \left(-\frac{\pi}{4}\right)}\right)^{12}$

$$= \left(\sqrt{2} \operatorname{cis} \frac{5\pi}{12}\right)^{12}$$

$$= 2^6 \operatorname{cis} 5\pi$$

$$= 64 (\cos 5\pi + i \sin 5\pi)$$

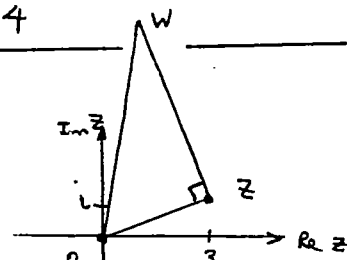
$$= 64 (-1 + 0i)$$

$$= -64$$

b) $z = 3 + i$

$$0 = 0 + 0i$$

$$w = ?$$



$2 \vec{zO}$ rotated clockwise $90^\circ = \vec{zW}$

$$2(0 - z)(-i) = w - z$$

$$2(-z)(-i) + z = w$$

$$2zi + z = w$$

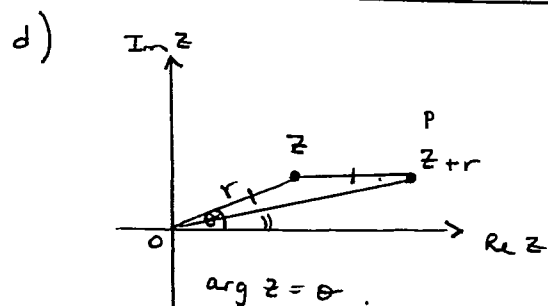
$$2(3+i)i + 3+i = w$$

$$6i - 2 + 3 + i = w$$

$$w = 1 + 7i$$

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c) $P : z = x + iy$
 $(x + iy)^2 = 2 - (x - iy)^2$
 $x^2 + 2xyi - y^2 = 2 - (x^2 - 2xyi - y^2)$
 $x^2 + 2xyi - y^2 = 2 - x^2 + 2xyi + y^2$
 $2x^2 - 2y^2 = 2$
 $x^2 - y^2 = 1$
 $\therefore$ rect. hyperbola.



$\angle OZP = 180 - \theta$ (co-interior angles supplementary on parallel lines).
 $\therefore \angle ZOP = \angle ZPO$ (angles opposite equal sides).
 $\therefore \angle ZOP = \frac{180 - (180 - \theta)}{2}$ (angle sum of triangle)
 $= \frac{180 - 180 + \theta}{2}$
 $= \frac{\theta}{2}$

$\therefore \arg(z+r) = \frac{1}{2}\theta$

OR $\arg(z+r) = \arg(re^{i\theta} + r)$
 $= \arg(r\cos\theta + r + ir\sin\theta)$
 $= \tan^{-1}\left[\frac{r\sin\theta}{r(\cos\theta + 1)}\right]$
 $= \tan^{-1}\frac{\sin\theta}{\cos\theta + 1}$

$\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$

and
 $\cos\theta = 2\cos^2\frac{\theta}{2} - 1$

So...

$\arg(z+r) = \tan^{-1}\left(\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2} - 1 + 1}\right)$
 $= \tan^{-1}\left(\frac{\cancel{2}\sin\frac{\theta}{2}\cancel{\cos\frac{\theta}{2}}}{\cancel{2}\cos^2\frac{\theta}{2}}\right)$
 $= \tan^{-1}\left(\tan\frac{\theta}{2}\right)$
 $= \frac{\theta}{2}$

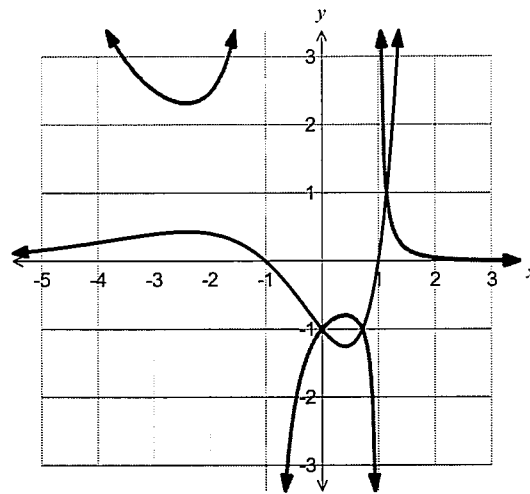
$\therefore \arg(z+r) = \frac{\theta}{2}$

Question 3 (15 marks)

Start a new sheet of writing paper.

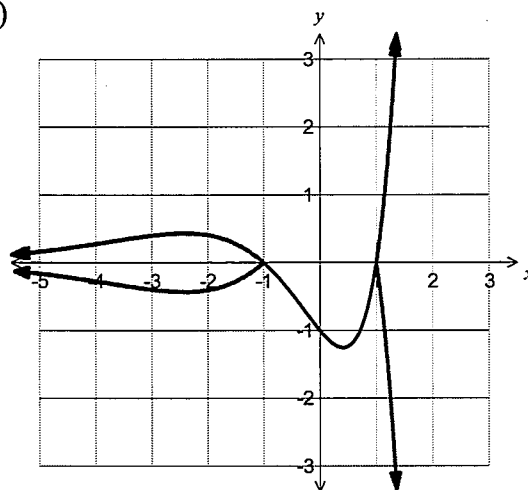
Marks

i) $y = \frac{1}{f(x)}$



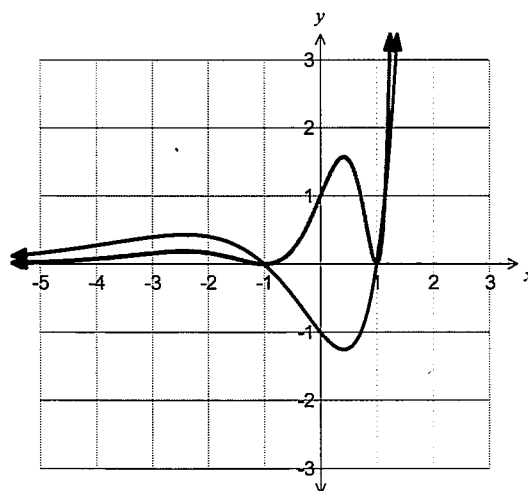
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ii) $|y| = f(x)$



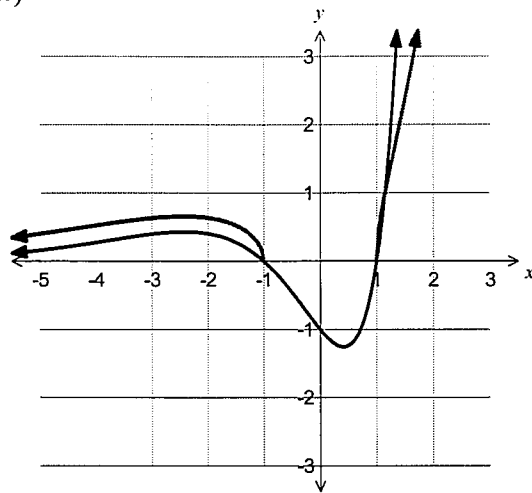
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iii) $y = [f(x)]^2$



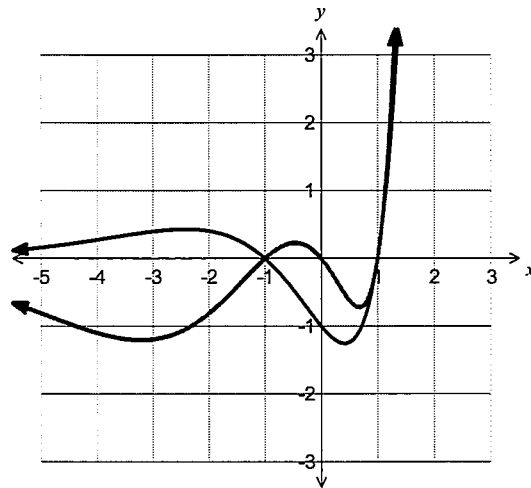
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iv) $y = \sqrt{f(x)}$



2

v) $y = x(f(x))$



2

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3 b) i) $1+i = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$

ii) $(1+i)^n = 2^{\frac{n}{2}} \operatorname{cis} \frac{\pi n}{4}$

$$(1-i)^n = 2^{\frac{n}{2}} \operatorname{cis} \left(-\frac{\pi n}{4}\right)$$

$$\therefore (1+i)^n + (1-i)^n = 2^{\frac{n}{2}} \operatorname{cis} \frac{\pi n}{4} + 2^{\frac{n}{2}} \operatorname{cis} \left(-\frac{\pi n}{4}\right)$$

$$= 2^{\frac{n}{2}} \left[\cos \frac{\pi n}{4} + i \sin \frac{\pi n}{4} + \cos \frac{\pi n}{4} - i \sin \frac{\pi n}{4} \right]$$

$$= 2^{\frac{n}{2}} \left[2 \cos \frac{\pi n}{4} \right]$$

$$= 2 \left[2^{\frac{n}{2}} \cos \frac{\pi n}{4} \right]$$

iii) $(1+x)^n = p_0 + p_1 x + p_2 x^2 + \dots + p_n x^n$

$$(1-x)^n = p_0 - p_1 x + p_2 x^2 - \dots + p_n x^n$$

$$(1+x)^n + (1-x)^n = 2p_0 + 2p_2 x^2 + 2p_4 x^4 + \dots + 2p_n x^n$$

Let $x = i$

$$(1+i)^n + (1-i)^n = 2p_0 + 2p_2(-1) + 2p_4(-1)^2 + \dots + 2p_n(-1)^{\frac{n}{2}}$$

$$2 \left[2^{\frac{n}{2}} \cos \frac{\pi n}{4} \right] = 2p_0 - 2p_2 + 2p_4 - \dots$$

$$2^{\frac{n}{2}} \cos \frac{\pi n}{4} = p_0 - p_2 + p_4 - \dots$$

Similarly,

$$(1+i)^n - (1-i)^n = 2i 2^{\frac{n}{2}} \sin \frac{\pi n}{4}$$

and

$$(1+x)^n - (1-x)^n = 2p_1 x + 2p_3 x^3 + \dots$$

Let $x = i$

$$(1+i)^n - (1-i)^n = 2p_1 i - 2p_3 i + 2p_5 i - \dots$$

$$2i 2^{\frac{n}{2}} \sin \frac{\pi n}{4} = 2ip_1 - 2ip_3 + 2ip_5 - \dots$$

$$\therefore 2^{\frac{n}{2}} \sin \frac{\pi n}{4} = p_1 - p_3 + p_5 - \dots$$

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Question 4:

$$a) i) \frac{(x-1)^2}{16} - \frac{(y+3)^2}{25} = 1$$

Centre $(1, -3)$

$$a = 4 \quad b = 5$$

$$e^2 = 1 + \left(\frac{5}{4}\right)^2$$

$$e^2 = 1 + \frac{25}{16}$$

$$e = \sqrt{\frac{41}{16}}, \quad e > 0$$

$$e = \frac{\sqrt{41}}{4}$$

$$\text{foci: } \begin{pmatrix} 1 + \sqrt{41} & , & -3 \\ 1 - \sqrt{41} & , & -3 \end{pmatrix}$$

ii) directrices:

$$x = 1 + \frac{16}{\sqrt{41}}, \quad x = 1 - \frac{16}{\sqrt{41}}$$

Asymptotes:

$$\frac{(x-1)^2}{16} - 1 = \frac{(y+3)^2}{25}$$

$$\frac{25}{16} (x-1)^2 \left[1 - \frac{16}{(x-1)^2} \right] = (y+3)^2$$

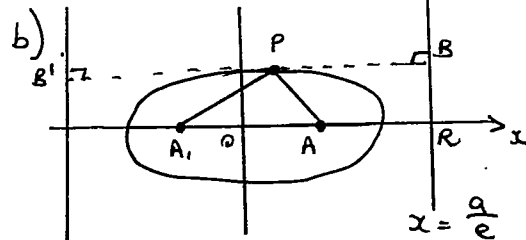
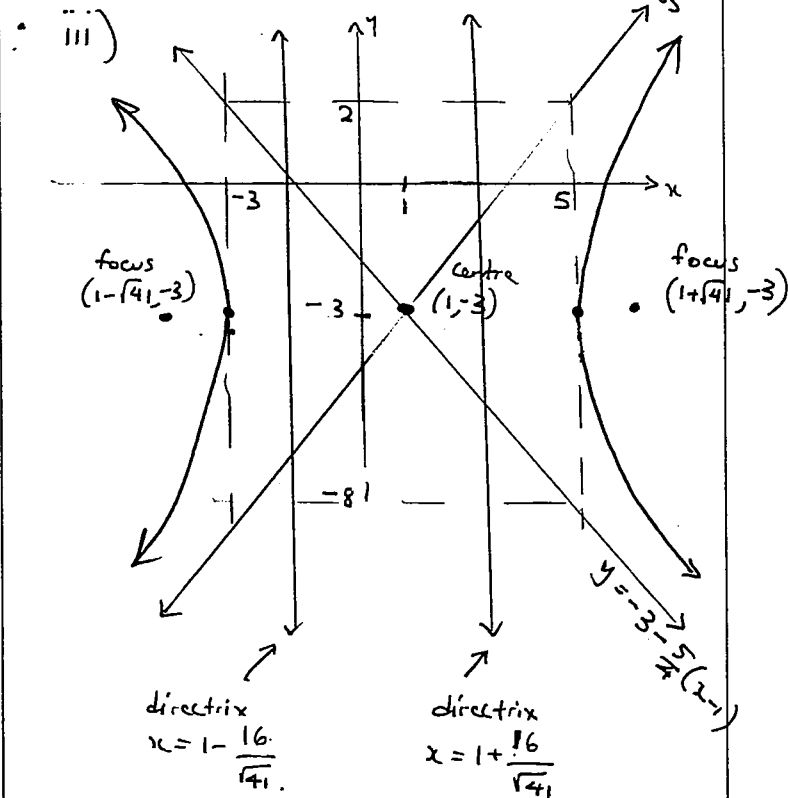
$$y+3 = \pm \frac{5}{4} (x-1) \sqrt{1 - \frac{16}{(x-1)^2}}$$

as $x \rightarrow \infty$

$$y+3 = \pm \frac{5}{4} (x-1)$$

 $\therefore$ asymptotes

$$y = -3 \pm \frac{5}{4} (x-1)$$



$$\frac{PA}{PB} = \frac{PA'}{PB'} = e$$

$$PA = e PB \quad PA' = e PB'$$

$$\therefore PA + PA' = e (PB + PB')$$

$$= e (2 \times OR)$$

$$= e \left(2 \times \frac{a}{e} \right)$$

$$= 2a$$

which is a constant.

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$$c) \quad x^2 - y^2 = a^2$$

$$i) \quad (a \sec \theta, a \tan \theta)$$

$$\begin{aligned} \text{LHS} &= (a \sec \theta)^2 - (a \tan \theta)^2 \\ &= a^2 (\sec^2 \theta - \tan^2 \theta) \\ &= a^2 (1) \\ &= a^2 \\ &= \text{RHS} \end{aligned}$$

$\therefore (a \sec \theta, a \tan \theta)$ are the parametric coordinates of a point on the hyperbola $x^2 - y^2 = a^2$

$$ii) \quad x^2 - y^2 = a^2$$

$$2x - 2y \frac{dy}{dx} = 0$$

$$2x = 2y \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{x}{y}$$

at $(a \sec \theta, a \tan \theta)$

$$\begin{aligned} m_{\text{tan}} &= \frac{a \sec \theta}{a \tan \theta} \\ &= \frac{1}{\cos \theta} \div \frac{\sin \theta}{\cos \theta} \\ &= \frac{1}{\sin \theta} \end{aligned}$$

$\therefore$ eqn

$$y - a \tan \theta = \frac{1}{\sin \theta} (x - a \sec \theta)$$

$$y - \frac{a \sin \theta}{\cos \theta} = \frac{1}{\sin \theta} \left(x - \frac{a}{\cos \theta} \right)$$

$$y - \frac{a \sin \theta}{\cos \theta} = \frac{x}{\sin \theta} - \frac{a}{\sin \theta \cos \theta}$$

$$\sin \theta \cos \theta y - a \sin^2 \theta = x \cos \theta - a$$

$$\begin{aligned} x \cos \theta - \sin \theta \cos \theta y &= a - a \sin^2 \theta \\ &= a(1 - \sin^2 \theta) \\ &= a \cos^2 \theta \end{aligned}$$

$$\therefore \div \cos \theta$$

$$x - \sin \theta y = a \cos \theta$$

iii) asymptotes of $x^2 - y^2 = a^2$ are $y = x$ & $y = -x$.

Consider $y = x$.

$$\therefore \text{Solve } x - \sin \theta y = a \cos \theta \text{ \& } y = x$$

$$x = a \cos \theta + x \sin \theta$$

$$x - x \sin \theta = a \cos \theta$$

$$x(1 - \sin \theta) = a \cos \theta$$

$$x = \frac{a \cos \theta}{1 - \sin \theta}$$

$$\text{Coord. of Point A.} \left\{ \begin{aligned} &\therefore y = \frac{a \cos \theta}{1 - \sin \theta} \quad (\text{since } y = x) \end{aligned} \right.$$

Consider $y = -x$

$$\therefore x = a \cos \theta + \sin \theta (-x)$$

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$$x + x \sin \theta = a \cos \theta$$

$$x(1 + \sin \theta) = a \cos \theta$$

coords
of point A

$$\left\{ \begin{array}{l} x = \frac{a \cos \theta}{1 + \sin \theta} \\ \therefore y = -\frac{a \cos \theta}{1 + \sin \theta} \quad (\text{since } y \equiv -x) \end{array} \right.$$

$$d_{OA} = \sqrt{\left(\frac{a \cos \theta}{1 - \sin \theta} - 0\right)^2 + \left(\frac{a \cos \theta}{1 - \sin \theta}\right)^2}$$

$$= \sqrt{2 \left(\frac{a \cos \theta}{1 - \sin \theta}\right)^2}$$

$$= \sqrt{2} \frac{(a \cos \theta)}{1 - \sin \theta}$$

similarly

$$d_{OB} = \sqrt{2} \frac{(a \cos \theta)}{1 + \sin \theta}$$

$$\therefore A = \frac{1}{2} b h$$

$$= \frac{1}{2} \sqrt{2} \frac{a \cos \theta}{1 - \sin \theta} \cdot \sqrt{2} \frac{a \cos \theta}{1 + \sin \theta}$$

$$= \frac{a^2 \cos^2 \theta}{1 - \sin^2 \theta}$$

$$= a^2 \frac{\cancel{\cos^2 \theta}}{\cancel{\cos^2 \theta}}$$

$$= a^2$$

which is a constant.

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Question 5

$$a) P(x) = x^4 - 6x^2 + 8x - 3$$

$$P(-1) = (-1)^4 - 6 - 8(-1) - 3$$

$$= 1 - 6 + 8 - 3$$

$$= 0$$

$$P'(x) = 4x^3 - 12x + 8$$

$$P'(-1) = -4 + 12 - 8$$

$$= 0$$

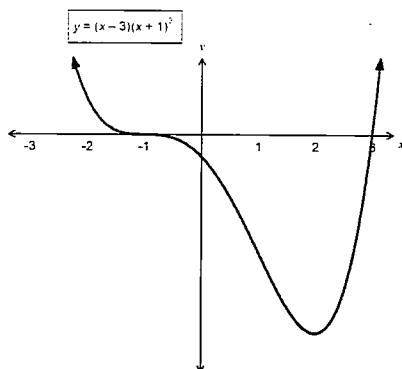
$$P''(x) = 12x^2 - 12$$

$$P''(-1) = 12 - 12$$

$$= 0$$

$$\therefore P(x) = (x+1)^3(x-a)$$

$$= (x+1)^3(x-3)$$



$$b) P(x) = ax^3 + bx^2 + cx + d$$

$$P(x) = k(x-5)(x+i)(x-i)$$

$$= k(x-5)(x^2+1)$$

$$= k(x^3 + x - 5x^2 - 5)$$

$$P(x) = k(x^3 - 5x^2 + x - 5)$$

$$P(2) = 15$$

$$P(2) = k(8 - 20 + 2 - 5)$$

$$15 = k(-15)$$

$$k = -1$$

$$\therefore P(x) = -x^3 + 5x^2 - x + 5$$

$$\therefore a = -1$$

$$b = 5$$

$$c = -1$$

$$d = 5$$

$$c) (i) (1-\sqrt{x})^{n-1} = (1-\sqrt{x})^n$$

$$= (1-\sqrt{x})^{n-1} (1 - (1-\sqrt{x}))$$

$$= (1-\sqrt{x})^{n-1} (1 - 1 + \sqrt{x})$$

$$= (1-\sqrt{x})^{n-1} \sqrt{x}$$

$$(ii) I_n = \int (1-\sqrt{x})^n dx$$

$$u = (1-\sqrt{x})^n$$

$$du = n(1-\sqrt{x})^{n-1} \cdot \frac{1}{2} x^{-1/2} dx$$

$$v = x$$

$$dv = 1$$

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$$= \left[x(1-\sqrt{x})^n \right]_0^1 + \frac{n}{2} \int_0^1 \frac{(1-\sqrt{x})^{n-1}}{\sqrt{x}} \cdot x \, dx$$

$$= [0] + \frac{n}{2} \int_0^1 (1-\sqrt{x})^{n-1} \cdot \sqrt{x} \, dx$$

$$= \frac{n}{2} \int_0^1 (1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n \, dx$$

$$= \frac{n}{2} I_{n-1} - \frac{n}{2} I_n$$

$$2I_n = n I_{n-1} - n I_n$$

$$(n+2) I_n = n I_{n-1}$$

$$I_n = \frac{n}{n+2} I_{n-1}$$

$$(iii) I_n = \frac{n}{n+2} I_{n-1}$$

$$= \frac{n}{n+2} \cdot \frac{n-1}{n+1} I_{n-2}$$

$$= \frac{n}{n+2} \cdot \frac{n-1}{n+1} \cdot \frac{n-2}{n} I_{n-3}$$

$$= \frac{n}{n+2} \cdot \frac{n-1}{n+1} \cdot \frac{n-2}{n} \dots \frac{2}{4} I_1$$

$$= \frac{n}{n+2} \cdot \frac{n-1}{n+1} \cdot \frac{n-2}{n} \times \dots \frac{2}{4} \times \frac{1}{3} I_0$$

$$= \frac{n! \cdot 2!}{(n+2)!}$$

$$\frac{1}{I_n} = \frac{(n+2)!}{n! \cdot 2!}$$

$$= {}^{n+2}C_2$$

$$d) \frac{a}{b} + \frac{b}{c} + \frac{c}{d} + \frac{d}{a}$$

$$\geq 2\sqrt{\frac{a}{b} \cdot \frac{b}{c}} + 2\sqrt{\frac{c}{d} \cdot \frac{d}{a}}$$

$$= 2\sqrt{\frac{a}{c}} + 2\sqrt{\frac{c}{a}}$$

$$= 2\left(\sqrt{\frac{a}{c}} + \sqrt{\frac{c}{a}}\right)$$

$$= 2\left(2\sqrt{\frac{a}{c} \cdot \frac{c}{a}}\right)$$

$$= 4(1)$$

$$\geq 4$$

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Question 6

$$\begin{aligned}
 (\cos \theta + i \sin \theta)^5 &= \cos 5\theta + i \sin 5\theta \\
 &= \cos^5 \theta + 5 \cos^4 \theta i \sin \theta - 10 \cos^3 \theta \sin^2 \theta \\
 &\quad - 10 \cos^2 \theta i \sin^3 \theta + 5 \cos \theta \sin^4 \theta \\
 &\quad + i \sin^5 \theta
 \end{aligned}$$

Equating imaginary:

$$\begin{aligned}
 \sin 5\theta &= 5 \cos^4 \theta \sin \theta - \\
 &\quad 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta \\
 &= 5(1 - \sin^2 \theta)^2 \sin \theta - \\
 &\quad 10(1 - \sin^2 \theta) \sin^3 \theta + \sin^5 \theta \\
 &= 5(1 - 2\sin^2 \theta + \sin^4 \theta) \sin \theta \\
 &\quad - 10 \sin^3 \theta + 10 \sin^5 \theta + \sin^5 \theta \\
 &= 5 \sin \theta - 10 \sin^3 \theta + 5 \sin^5 \theta \\
 &\quad - 10 \sin^3 \theta + 10 \sin^5 \theta + \sin^5 \theta \\
 &= 16 \sin^5 \theta - 20 \sin^3 \theta + 5 \sin \theta
 \end{aligned}$$

(ii) Let $x = \sin \theta$

$$16x^5 - 20x^3 + 5x = 0$$

$$x(16x^4 - 20x^2 + 5) = 0$$

$$\sin 5\theta = 0$$

$$5\theta = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$\theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}$$

$$\text{So } 16x^4 - 20x^2 + 5 = 0$$

$$x = \sin \frac{\pi}{5}, \sin \frac{2\pi}{5}, \sin \frac{3\pi}{5}, \sin \frac{4\pi}{5}$$

$$\therefore x = \sin \frac{k\pi}{5}, k = 1, 2, 3, 4$$

$$(iii) y = x + 1$$

$$\therefore x = y - 1$$

$$16(y-1)^4 - 20(y-1)^2 + 5 = 0$$

$$\begin{aligned}
 16(y^4 - 4y^3 + 6y^2 - 4y + 1) \\
 - 20(y^2 - 2y + 1) + 5 = 0
 \end{aligned}$$

$$\begin{aligned}
 16y^4 - 64y^3 + 96y^2 - 64y + 16 \\
 - 20y^2 + 40y - 20 + 5 = 0
 \end{aligned}$$

$$16y^4 - 64y^3 + 76y^2 - 24y + 1 = 0$$

$$\therefore 16x^4 - 64x^3 + 76x^2 - 24x + 1 = 0$$

(iv) For (iii)

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$$

= roots 3 at a time
 roots 4 at a time

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$$= \frac{24}{16} \div \frac{1}{16}$$

$$= \frac{24}{16} \times \frac{16}{1}$$

$$= 24$$

$$b) 5x^2 - 6xy + y^2 - 2x + 4y - 3 = 0$$

$$10x - 6x \frac{dy}{dx} + y(-6) + 2y \frac{dy}{dx} - 2$$

$$+ 4 \frac{dy}{dx} = 0$$

$$(4 - 6x + 2y) \frac{dy}{dx} = 6y + 2 - 10x$$

$$\frac{dy}{dx} = \frac{6y + 2 - 10x}{4 - 6x + 2y}$$

at (1, 2)

$$\frac{dy}{dx} = \frac{6(2) + 2 - 10}{4 - 6 + 2(2)}$$

$$= \frac{4}{2}$$

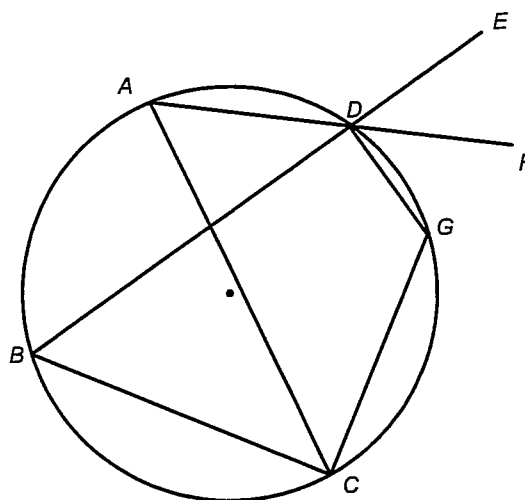
$$= 2$$

$$y - 2 = 2(x - 1)$$

$$y - 2 = 2x - 2$$

$$y = 2x$$

c).



$$\text{Let } \angle ACG = x$$

$$\therefore \angle BCA = x \text{ (AC bisects } \angle BCG \text{)}$$

$$\therefore \angle ADB = x \text{ (angles standing on same arc are equal)}$$

$$\therefore \angle EDF = x \text{ (vertically opposite angles equal)}$$

$$\begin{aligned} \angle BDG &= 180 - \angle BCG \text{ (opposite angles in cyclic quad are supplementary)} \\ &= 180 - 2x \end{aligned}$$

$$\begin{aligned} \therefore \angle FDG &= 180 - \angle BDG - \angle ADB \\ &= 180 - (180 - 2x) - x \text{ (angle sum of straight line)} \\ &= 2x - x \\ &= x \end{aligned}$$

$$\therefore \angle EDF = \angle FDG$$

$$\therefore FD \text{ bisects } \angle EDG$$

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Question 7

$$a) \frac{d}{dx}(x^n) = n x^{n-1}$$

$$\frac{d^2}{dx^2}(x^n) = \frac{d}{dx}(n x^{n-1})$$

$$= n(n-1) x^{n-2}$$

$$\frac{d^3}{dx^3}(x^n) = n(n-1)(n-2) x^{n-3}$$

$$\therefore \frac{d^n}{dx^n} x^n = n(n-1)(n-2)\dots(n-(n+1)) x^0$$

$$= n(n-1)(n-2)\dots(1) x^0$$

$$= n!$$

$$ii) \frac{d^n}{dx^n}(x^n \ln x) = n! \left(\ln x + \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{n} \right)$$

For $n=1$

$$\frac{d}{dx}(x \ln x) = x \cdot \frac{1}{x} + 1 \cdot \ln x$$

$$= 1 + \ln x$$

$$= 1!(1 + \ln x)$$

Assume true for $n=k$

i.e.

$$\frac{d^k}{dx^k}(x^k \ln x)$$

$$= k! \left(\ln x + \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{k} \right)$$

$$\frac{d^{k+1}}{dx^{k+1}}(x^{k+1} \ln x)$$

$$= \frac{d}{dx} \left(\frac{d^k}{dx^k}(x^{k+1} \ln x) \right)$$

$$= \frac{d^k}{dx^k} \left(\frac{d}{dx}(x^{k+1} \ln x) \right)$$

$$= \frac{d^k}{dx^k} \left((k+1)x^k \ln x + x^{k+1} \cdot \frac{1}{x} \right)$$

$$= \frac{d^k}{dx^k} \left((k+1)x^k \ln x + x^k \right)$$

$$= (k+1) \left[\frac{d^k}{dx^k}(x^k \ln x) + \frac{d^k}{dx^k}(x^k) \right]$$

$$+ \frac{d^k}{dx^k}(x^k)$$

$$= (k+1) k! \left(\ln x + \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{k} \right)$$

$$+ k!$$

$$= (k+1) k! \left(\ln x + \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{k} + \frac{1}{k+1} \right)$$

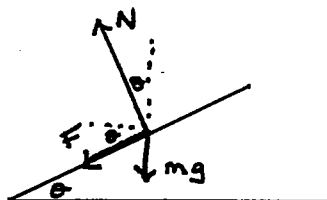
$$= (k+1)! \left(\ln x + \frac{1}{1} + \frac{1}{2} + \dots + \frac{1}{k} + \frac{1}{k+1} \right)$$

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7b) $r = 500 \text{ m}$



i)



c) $+\uparrow \quad \downarrow mg \quad \downarrow R$

i) $m\ddot{x} = -mg - kmv$

$$\ddot{x} = -g - kv$$

$$\ddot{x} = -(g + kv)$$

ii) vert: $N \cos \theta - F \sin \theta - mg = 0$

horiz: $N \sin \theta + F \cos \theta = \frac{mv^2}{r}$

if no sideways force $F = 0$

$$\therefore N \cos \theta = mg$$

$$N \sin \theta = \frac{mv^2}{r}$$

$$\therefore \tan \theta = \frac{mv^2}{r} \div mg$$

$$= \frac{mv^2}{r} \times \frac{1}{mg}$$

$$\therefore \tan \theta = \frac{v^2}{rg}$$

ii) $\frac{dv}{dt} = -(g + kv)$

$$\frac{dt}{dv} = -\frac{1}{g + kv}$$

$$t = \int -\frac{1}{g + kv} dv$$

$$t = -\int \frac{1}{g + kv} dv$$

$$t = -\left[\frac{1}{k} \int \frac{k}{g + kv} dv \right]$$

$$t = -\frac{1}{k} \ln(g + kv) + c$$

when $t = 0$ $v = u$.

$$\therefore 0 = -\frac{1}{k} \ln(g + ku) + c$$

$$c = \frac{1}{k} \ln(g + ku)$$

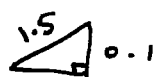
$$\therefore t = \frac{1}{k} \ln(g + ku) - \frac{1}{k} \ln(g + kv)$$

at max height $v = 0$, $t = T$

$$\therefore T = \frac{1}{k} \ln(g + ku) - \frac{1}{k} \ln(g)$$

$$\therefore T = \frac{1}{k} \ln\left(\frac{g + ku}{g}\right)$$

iii) we know



by pyth

$$x = 1.496...$$

$$\therefore \tan \theta = \frac{0.1}{1.496...}$$

$$\therefore v^2 = rg \left(\frac{0.1}{1.496...} \right)$$

$$v = 18.1 \text{ m/s}$$

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$$\text{iii) } \ddot{x} = -(g + kv)$$

$$v \frac{dv}{dx} = -(g + kv)$$

$$\frac{dv}{dx} = -\frac{(g + kv)}{v}$$

$$\frac{dx}{dv} = -\frac{v}{g + kv}$$

$$x = -\int \frac{v}{g + kv} dv$$

$$x = -\int \left(\frac{1}{k} - \frac{\frac{1}{k}}{kv + g} \right) dv$$

$$x = -\int \left(\frac{1}{k} - \frac{g}{k} \frac{1}{kv + g} \right) dv$$

$$x = -\int \left(\frac{1}{k} - \frac{g}{k^2} \frac{k}{kv + g} \right) dv$$

$$x = -\left[\frac{1}{k} v - \frac{g}{k^2} \ln(kv + g) \right] + C$$

$$\text{when } v = u \quad x = 0$$

$$0 = -\left[\frac{u}{k} - \frac{g}{k^2} \ln(ku + g) \right] + C$$

$$C = \frac{u}{k} - \frac{g}{k^2} \ln(g + ku)$$

$$\therefore x = -\frac{1}{k} v + \frac{g}{k^2} \ln(kv + g) + \frac{u}{k} - \frac{g}{k^2} \ln(g + ku)$$

$$\text{at max } v = 0, x = h$$

$$\therefore h = 0 + \frac{g}{k^2} \ln g + \frac{u}{k} - \frac{g}{k^2} \ln(g + ku)$$

$$h = \frac{1}{k} \left[u + \frac{g}{k} \ln \left(\frac{g}{g + ku} \right) \right]$$

$$= \frac{1}{k} \left[u + g(-T) \right]$$

$$\text{Since } TK = \ln \left(\frac{g + ku}{g} \right)$$

$$-TK = -\ln \left(\frac{g + ku}{g} \right)$$

$$-TK = \ln \left(\frac{g}{g + ku} \right)$$

$$-T = \frac{1}{k} \ln \left(\frac{g}{g + ku} \right)$$

$$\therefore h = \frac{1}{k} [u - gT]$$

Question 8:

$$\text{a) i) } \ddot{y} = -g$$

$$\dot{y} = -g \int dt$$

$$\dot{y} = -gt + c$$

$$u \sin \alpha = 0 + c_1$$

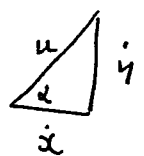
$$\therefore \dot{y} = -gt + u \sin \alpha$$

$$y = -\frac{1}{2}gt^2 + ut \sin \alpha + c_2$$

$$0 = 0 + 0 + c_2$$

$$\therefore y = -\frac{1}{2}gt^2 + ut \sin \alpha$$

Initial Conditions



$$\dot{x} = u \cos \alpha$$

$$\dot{y} = u \sin \alpha$$

$$t = 0$$

$$x = 0$$

$$y = 0$$

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$$\ddot{x} = 0$$

$$\dot{x} = c_3$$

$$u \cos \alpha = c_3$$

$$\therefore \dot{x} = u \cos \alpha$$

$$x = ut \cos \alpha + c_4$$

$$0 = 0 + c_4$$

$$\therefore x = ut \cos \alpha$$

Cartesian equation:

$$t = \frac{x}{u \cos \alpha}$$

$$y = -\frac{1}{2}g \left(\frac{x}{u \cos \alpha} \right)^2 + u \left(\frac{x}{u \cos \alpha} \right) \sin \alpha$$

$$y = -\frac{1}{2}g \frac{x^2}{u^2 \cos^2 \alpha} + x \tan \alpha$$

$$y = -\frac{gx^2}{2u^2} \sec^2 \alpha + x \tan \alpha$$

ii) (a, b) satisfies

$$b = -\frac{ga^2}{2u^2} (\tan^2 \alpha + 1) + a \tan \alpha$$

$$2u^2 b = -ga^2 \tan^2 \alpha - ga^2 + 2u^2 a \tan \alpha$$

$$ga^2 \tan^2 \alpha - 2u^2 a \tan \alpha + (2u^2 b + ga^2) = 0$$

if 2 trajectories $\Delta > 0$

$$b^2 - 4ac > 0$$

$$(-2u^2 a)^2 - 4(ga^2)(2u^2 b + ga^2) > 0$$

$$4u^4 a^2 - 8ga^2 bu^2 - 4g^2 a^4 > 0$$

$$\div 4a^2$$

$$u^4 - 2gbu^2 - g^2 a^2 > 0$$

$$\left[(u^2 - gb)^2 = u^4 - 2u^2 gb + g^2 b^2 \right]$$

$$\therefore (u^2 - gb)^2 - g^2 a^2 - g^2 b^2 > 0$$

$$(u^2 - gb)^2 > g^2 a^2 + g^2 b^2$$

$$(u^2 - gb)^2 > g^2 (a^2 + b^2)$$

b) i) Vert: $T \cos \alpha + N \sin \alpha - mg = 0$

$$T \cos \alpha + N \sin \alpha = mg \quad (1)$$

horiz: $T \sin \alpha - N \cos \alpha = mr\omega^2$

$$\therefore \cos \frac{\pi}{3} = \frac{1}{2} \quad \sin \frac{\pi}{3} = \frac{\sqrt{3}}{2}$$

$$\therefore T\left(\frac{1}{2}\right) + N\left(\frac{\sqrt{3}}{2}\right) = mg$$

$$T + \sqrt{3}N = 2mg \quad (1)$$

$$T\left(\frac{\sqrt{3}}{2}\right) - N\left(\frac{1}{2}\right) = mr\omega^2$$

$$\sqrt{3}T - N = 2mr\omega^2 \quad (2)$$

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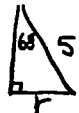
$$\text{ii) } N = \sqrt{3}T - 2mr\omega^2$$

$$\therefore T + \sqrt{3} [\sqrt{3}T - 2mr\omega^2] = 2mg$$

$$T + 3T - 2\sqrt{3}mr\omega^2 = 2mg$$

$$4T = 2mg + 2\sqrt{3}mr\omega^2$$

$$T = \frac{m}{4} [2g + 2\sqrt{3}r\omega^2]$$

we know  $\therefore \sin 60 = \frac{r}{5}$

$$r = \frac{5\sqrt{3}}{2}$$

$$\therefore T = \frac{m}{4} \left[2g + 2\sqrt{3} \cdot \frac{5\sqrt{3}}{2} \omega^2 \right]$$

$$\therefore T = \frac{m}{4} [2g + 15\omega^2]$$

$$N = \sqrt{3}T - 2mr\omega^2$$

$$= \sqrt{3} \left[\frac{m}{4} (2g + 15\omega^2) \right] - 2m \frac{5\sqrt{3}}{2} \omega^2$$

$$= \sqrt{3} \left[\frac{2mg}{4} + \frac{15m\omega^2}{4} \right] - 5\sqrt{3}m\omega^2$$

$$= \frac{\sqrt{3}}{4} [2mg + 15m\omega^2 - 20m\omega^2]$$

$$\therefore N = \frac{\sqrt{3}}{4} m (2g - 5\omega^2)$$

$$\text{iii) need } N > 0$$

$$\frac{2\sqrt{3}mg}{4} - \frac{5\sqrt{3}}{4}m\omega^2 > 0$$

$$2\sqrt{3}g - 5\sqrt{3}\omega^2 > 0$$

$$5\sqrt{3}\omega^2 - 2\sqrt{3}g < 0$$

$$\therefore \omega^2 < \frac{2\sqrt{3}g}{5\sqrt{3}}$$

$$\therefore \omega^2 < \frac{2g}{5}$$

end of exam